

# Custodial implementation: reference map

The equation each formula comes from, and the figure each paper computes.

Notation:  $L = k\pi r_c \approx 35$  (warped volume);  $M_{KK}$  first KK mass;  $v \approx 174$  GeV;  $x_1 \approx 2.45$  (first gauge-KK root);  $B_d, B_Q$  flavour-sum factors (CGHNP);  $s = \sin \theta_W$ ,  $c_W = \cos \theta_W$ .

## Formula sources

### Flavor Physics in the Randall-Sundrum Model, I

Casagrande, Goertz, Haisch, Neubert, Pfoh (CGHNP)

arXiv:0807.4937

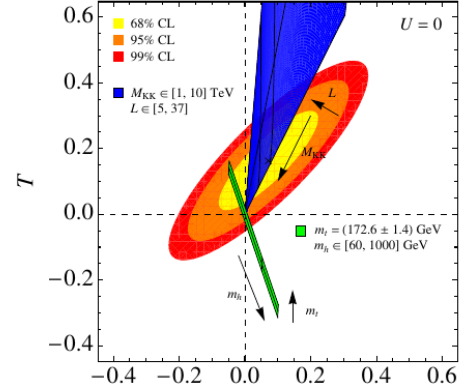


Fig. 4: minimal-RS  $S$ - $T$  wedge, the volume-enhanced  $T$  that custodial  $SU(2)_R$  then protects.

Formulas taken

$$\Delta T_{\min} = \frac{\pi v^2}{2c_W^2 M_{KK}^2} \left( L - \frac{1}{2L} \right) \quad (\text{Eq. 147})$$

$$\Delta T_{\text{cust}} = -\frac{\pi v^2}{4c_W^2 M_{KK}^2} \frac{1}{L} \quad (\text{Eq. 153, after ADMS})$$

$$\delta g_L^b = +\frac{m_b^2}{2M_{KK}^2} B_d, \quad \delta g_R^b = -\frac{m_b^2}{2M_{KK}^2} B_Q \quad (\text{Eq. 170})$$

### RS1, Custodial Isospin and Precision Tests

Agashe, Delgado, May, Sundrum (ADMS)

hep-ph/0308036

Formula taken (mechanism)

No closed form is copied.  $SU(2)_R$  removes the  $\log(k\pi r_c)$  enhancement of  $T$ :

$$T : \underbrace{+\frac{\pi L}{2c_W^2}}_{\text{minimal}} \xrightarrow{SU(2)_R} \underbrace{-\frac{\pi}{4c_W^2 L}}_{\text{custodial}} \quad (\text{ADMS Eq. 4.9/5.6})$$

the  $+L \rightarrow -1/L$  flip that CGHNP distil into Eq. 153.

What the paper computes

$T$  (protected, leaving a  $\sim 3$  TeV floor set by  $S$ ),  $S$  (unprotected), and  $Zb_L$ . Analytic paper; no data figure.

### A Custodial Symmetry for $Zb\bar{b}$

Agashe, Contino, Da Rold, Pomarol (ACDP)

hep-ph/0605341

Formulas taken

$$P_{LR} : c_1 = c_2 \Rightarrow \delta g_L^b|_{\text{tree}} = 0, \quad q_L \in (2, 2)_{2/3} \quad (\text{Eqs. 3-12})$$

$$b_R \in (1, 1)_{-1/3} : T_L^3 = T_R^3 = 0 \Rightarrow \delta g_R^b = 0 (P_C) (\text{Eq. 22 / Table 1})$$

What the paper computes

$\delta g_L^b$  (protected) and the representation-dependent  $\delta g_R^b$ , custodial versus minimal; the  $A_{FB}^b$  anomaly. No data figure (analytic).

### Light Kaluza-Klein States, and Electroweak Constraints, with Custodial $SU(2)$

Carena, Pontón, Santiago, Wagner (CPSW)

hep-ph/0607106, hep-ph/0701055

Formulas taken (top-partner one loop)

$$\delta g_L^{b, \text{singlet}} = \frac{\alpha}{16\pi s^2 M_W^2} \frac{m_{q_0 t}^4}{M_t^2} [1 + \dots] \quad (\text{Eq. 14})$$

$$\delta g_L^{b, \text{bidoublet}} = \frac{\alpha}{32\pi s^2 M_W^2} m_t^2 [\dots] \quad (\text{Eq. 15})$$

$$\Delta T_{\text{top}} = T_{\text{top}} \frac{2m_{q_0 t}^2}{M_t^2} [\dots] \quad (\text{Eq. 13, } T_{\text{top}} \text{ from Eq. 35})$$

## Comparison and validation (no formulas taken)

### The Flavor of the Composite Pseudo-Goldstone Higgs

Csáki, Falkowski, Weiler (CFW)

arXiv:0804.1954

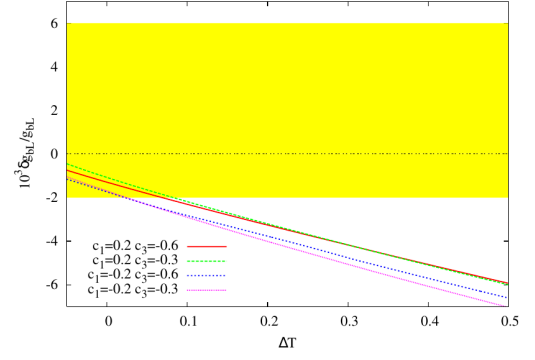


Fig. 1: loop  $\delta g_{bL}/g$  vs.  $\Delta T$ . Top partners re-introduce a negative  $\delta g_L^b$  correlated with  $+\Delta T$  (bidoublet  $-T$ , singlet  $+T$ ); floor  $\sim 2.5\text{--}3$  TeV ( $S$ -set).

Formulas taken none.

What the paper computes

Anarchic  $\varepsilon_K$  from the KK gluon ( $\text{Im } \Lambda_{LR}$ ,  $s \rightarrow d$ ):

$M_{KK} \gtrsim 21$  TeV (RS),  $33$  TeV (gauge-Higgs),

with the electroweak sector held near 3 TeV by custodial. The explicit electroweak-versus-flavor split we reproduce.

### $\Delta F = 2$ Observables and Fine-Tuning with Custodial Protection

Blanke, Buras, Duling, Gori, Weiler (Blanke et al.)

arXiv:0809.1073

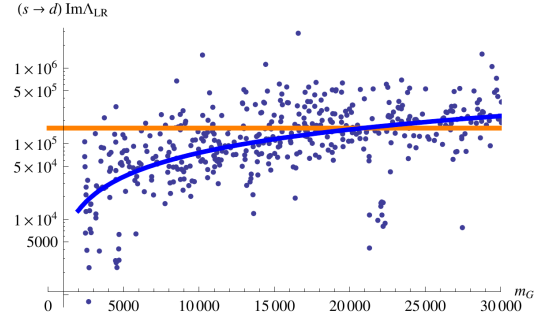


Fig. 1: KK-gluon  $\text{Im } \Lambda_{LR}$  vs.  $m_G$ ; the bound is crossed around  $m_G \approx 20$  TeV.

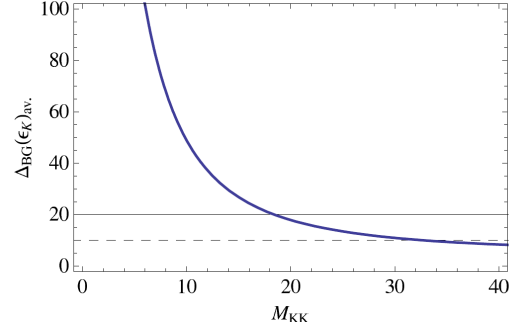


Fig. 9:  $\varepsilon_K$  fine-tuning  $\Delta_{BG}$  vs.  $M_{KK}$ , computed in the full custodial model. The wall survives custodialization.

*Formulas taken none.*

*What the paper computes*

The full  $\Delta F = 2$  set *inside* the custodial model. The  $\varepsilon_K$  fine-tuning  $\Delta_{BG}$  vs.  $M_{KK}$  gives

$$M_{KK} \gtrsim 18 \text{ TeV } (\Delta_{BG} < 20), \quad 30 \text{ TeV } (\Delta_{BG} < 10).$$

Direct evidence that custodial does not move the flavor bound.

## Flavor Physics in the Randall-Sundrum Model, II

Bauer, Casagrande, Haisch, Neubert (Bauer et al., II)

arXiv:0912.1625

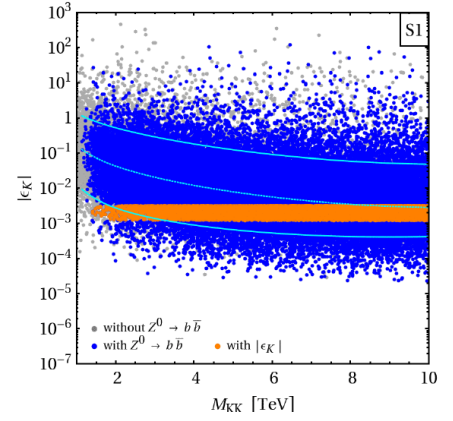


Fig. 4: anarchic  $|\varepsilon_K|$  vs.  $M_{KK}$  (S1), spanning  $\sim 6$  decades; the 95% floor sits near 10 TeV.

*Formulas taken none.*

*What the paper computes*

The anarchic minimal  $|\varepsilon_K|$  cloud (our Lane A baseline), 95%-quantile floor

$$M_g^{(1)} \gtrsim 10 \text{ TeV (S1)};$$

$C_4$  is “a pure QCD effect ... insensitive to the electroweak embedding”; notes  $Zb_L$  is removable by custodial.

Formulas come from CGHNP, ADMS, ACDP, and CPSW. CFW, Blanke, and Bauer II supply the comparison figures: custodial protection lowers the electroweak floors for  $S, T, U$  and  $Z \rightarrow b\bar{b}$  to a few TeV, while the  $\varepsilon_K$  wall stays at tens of TeV. Full equation-level provenance (status tags, page numbers, the four proxy gaps) is in `docs/CUSTODIAL_PROVENANCE.md`.